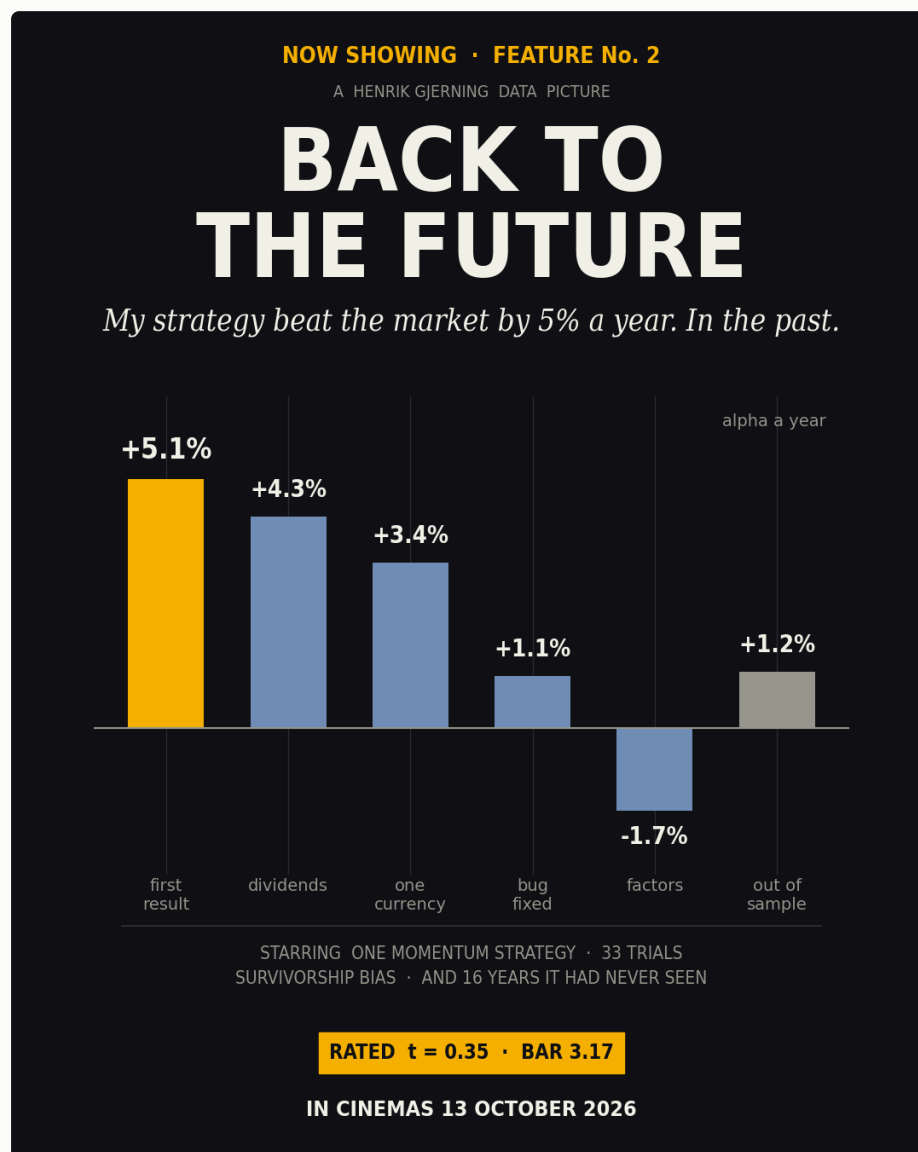


# Back to the Future: my strategy beat the market by 5% a year. In the past.

One momentum strategy, six honest corrections, and what was left



Henrik Gjernerig · Rude Investment Consulting · September 2026 · Project 10, Case 33

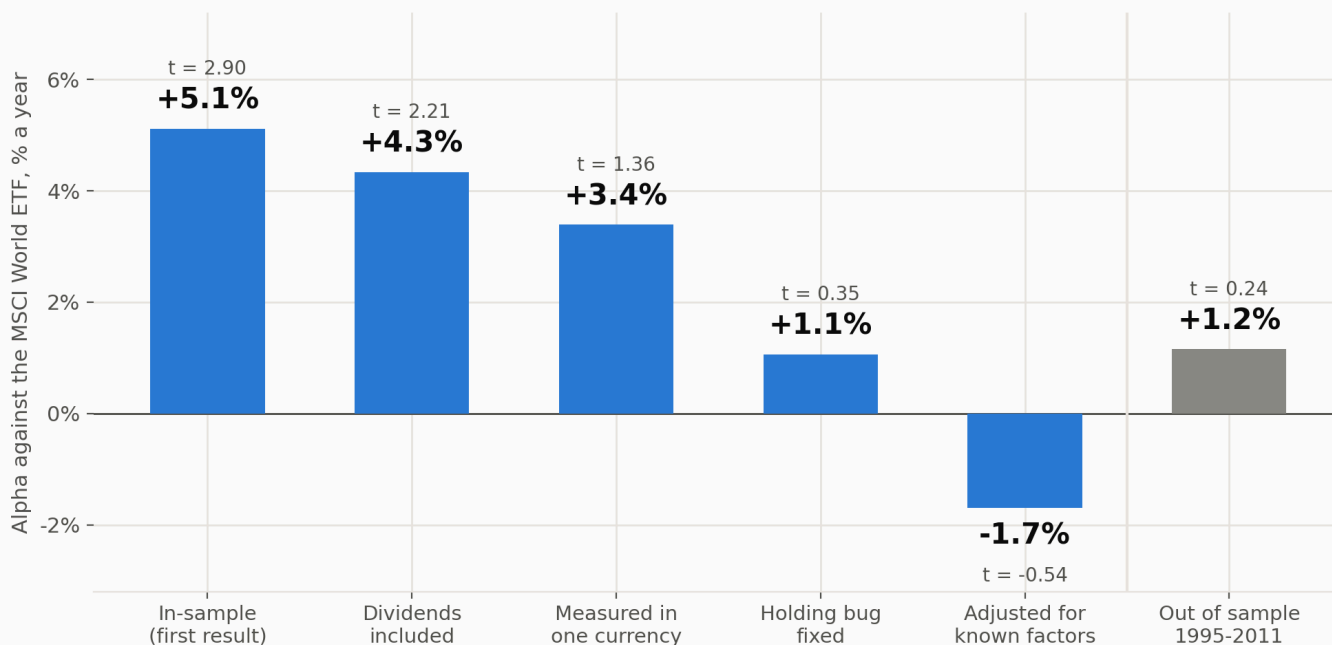
**In one paragraph.** On 1 September 2026 a weekly momentum strategy I had built showed **+5.1% a year of alpha** against the MSCI World ETF, with a t-statistic of 2.90. Two weeks of checking later, the same strategy's number of record is **+1.1% ( $t$  0.35)**. Adjusted for known factors it is **-1.7%**. On sixteen years of point-in-time data it had never seen, it earned **+1.2% ( $t$  0.24)** with a -79% drawdown against the market's -56%. Nothing was fraudulent and nothing was unusual. Every correction was a standard one: dividends, currency, a code bug, survivorship, the right null hypothesis, and counting the trials. The haircut, 79%, is in line with what the

literature finds for published anomalies and for bank strategies once they go live. The only result that survived was about **trading less**.

**Statistics box.** In-sample: 764 weeks, 2012-01-20 to 2026-09-04. Number of record: alpha +1.06% a year, 95% CI [-4.8%, +7.1%], t 0.35 (block bootstrap) / 0.33 (OLS), beta 0.85. Trials 33, Bonferroni bar t 3.17. Harvey–Liu–Zhu haircut of the first t at 33 trials: 47%. Out of sample: 885 weeks, alpha +1.15%, t 0.24. Cross-sectional placebo z: 4.92 on today's members, 1.34 on point-in-time members.

## My strategy beat the market by 5% a year. Then I tested it properly.

The same strategy, restated after each correction. A t-stat of 3.17 was needed after 33 trials. None got there.



Weekly returns 2012-2026 (in-sample) and 1995-2011 (out of sample, point-in-time universe). Henrik Gjering

## 1. The result that looked too good

The strategy is a long-only, weekly-rebalanced momentum book. Each week it ranks large-cap stocks in the US and Europe on eight price-only features (12- and 6-month momentum, trend, volatility, short-term reversal and a few more) and holds the ten best per region, weighted by inverse volatility. It pays a measured, realistic commission and prefers names it already holds, to save turnover.

The first regression against the MSCI World ETF, over 2012–2026, gave +5.11% a year, t 2.90, and a Sharpe ratio of 0.945 against the benchmark's 0.695. That is the result a quant hopes for. It is also exactly what the literature warns about. The rest of this article is what happened when I tried to break it.

## 2. The waterfall

**Table 1. The same strategy, restated after each correction**

| #   | Date (2026) | Restatement                               | What changed                                                                                                    | Alpha / yr    | t (type)               | Sharpe: strategy vs benchmark |
|-----|-------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------|------------------------|-------------------------------|
| 1   | 09-01       | In-sample (price basis)                   | First regression alpha vs MSCI World ETF; non-US legs and benchmark on price-only closes                        | <b>+5.11%</b> | 2.90 (block bootstrap) | 0.945 vs 0.695                |
| 2   | 09-08       | Total-return re-basing                    | All legs and the benchmark moved to total return (dividends)                                                    | <b>+4.34%</b> | 2.21 (block bootstrap) | 0.932 vs 0.819                |
| 3   | 09-11       | Stated in USD                             | EU legs had been summed in local currencies; restated in the benchmark's currency (measured on the step-2 book) | <b>+3.39%</b> | 1.36 (OLS)             | 0.874 vs 0.819                |
| 4   | 09-11       | Holding-lock bug fixed (number of record) | A holding bonus had frozen the book into ~40 names held for years; reverted on mechanics                        | <b>+1.06%</b> | 0.35 (block bootstrap) | 0.663 vs 0.819                |
| 5   | 09-12       | Factor-adjusted (13 JKP themes)           | Momentum loading +0.51 (t 5.8) explains the rest                                                                | <b>-1.69%</b> | -0.54 (OLS)            | n/a                           |
| OOS | 09-01       | Out of sample, 1995–2011                  | Same rules on a point-in-time universe never used in development                                                | <b>+1.15%</b> | 0.24                   | 0.475 vs 0.540                |

*The rows are restatements in the order they happened, not additive components: row 3 was measured on the row-2 book, and row 4 is today's number of record. Sources for every row are in `data/waterfall_sources.csv`.*

### Correction 1: dividends (+5.1% → +4.3%)

The US leg used total-return prices. The European legs, and the benchmark itself, used price-only closes. Mixing the two quietly favours whichever side leaves out dividends, and the benchmark's return rose more when dividends were added: its Sharpe went from 0.695 to 0.819. **Lesson: strategy and benchmark must be on the same return basis, every leg.**

### Correction 2: currency (+4.3% → +3.4%)

European legs were summed in their local currencies (EUR, SEK, DKK, CHF and PLN) as if they had been hedged for free, while the benchmark is an unhedged USD fund. Restated in the benchmark's

currency, the alpha fell to +3.4% and the t to 1.36. **Lesson: a return has a currency. Say which one.**

### Correction 3: the holding bug (→ +1.1%)

To save trading costs, the strategy gives a bonus to stocks it already holds. A parameter search had set that bonus so high that the book barely traded: about forty names over fifteen years, changed in roughly one week in seventeen. It had become a buy-and-hold portfolio of stocks picked early in the sample. Resetting the bonus on mechanical grounds, not on performance, gives today's number of record: **+1.06% a year, t 0.35**. Its Sharpe of 0.663 is now *below* the benchmark's 0.819. Maximum drawdown is -37.6% against -32.7%. I re-derived these from the stored weekly series and they match to the last digit.

### Correction 4: known factors (→ -1.7%)

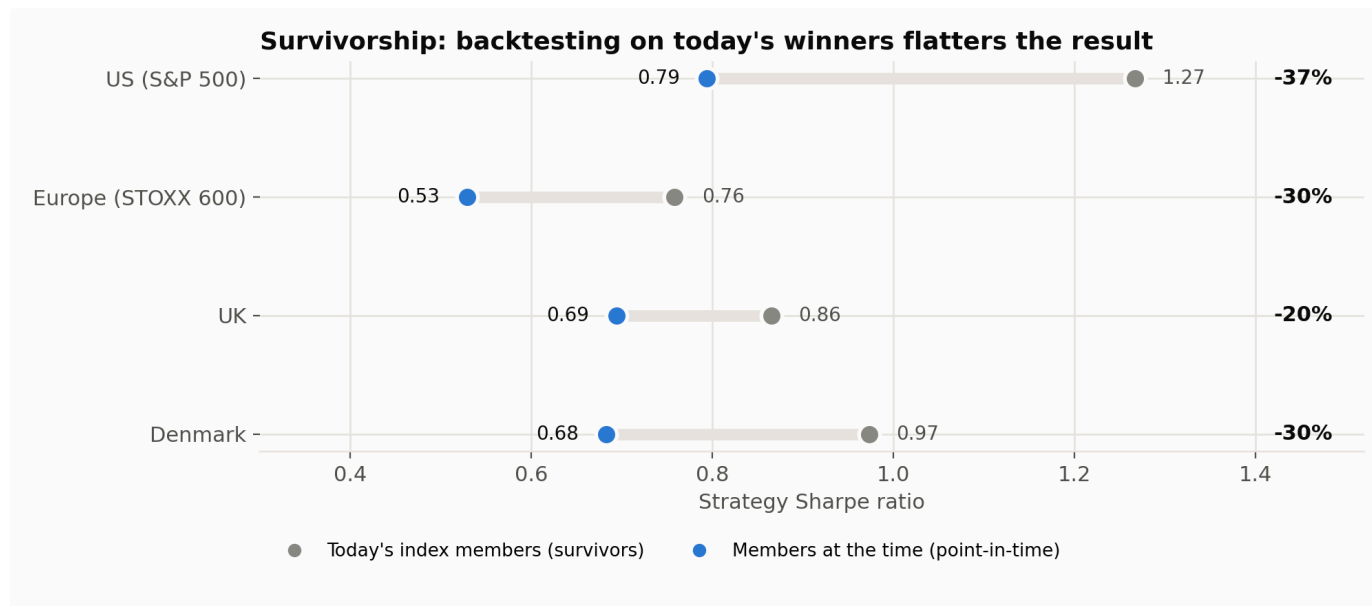
With the bug fixed, the strategy's momentum loading appears clearly (+0.51, t 5.8, on 13 JKP themes). Regressed on those factors, the alpha is -1.7% (t -0.54). Across four attribution models it lies between -1.7% and +2.9% a year, and every t is below 1. **It is a momentum fund, and momentum is available cheaply.**

## 3. Survivorship: backtesting on today's winners

The easiest universe to download is today's index. It is also the most dangerous. Every stock in it survived to be there, and a momentum strategy is especially good at finding the ones that went on to survive.

**Table 2. The same strategy on today's members vs members at the time**

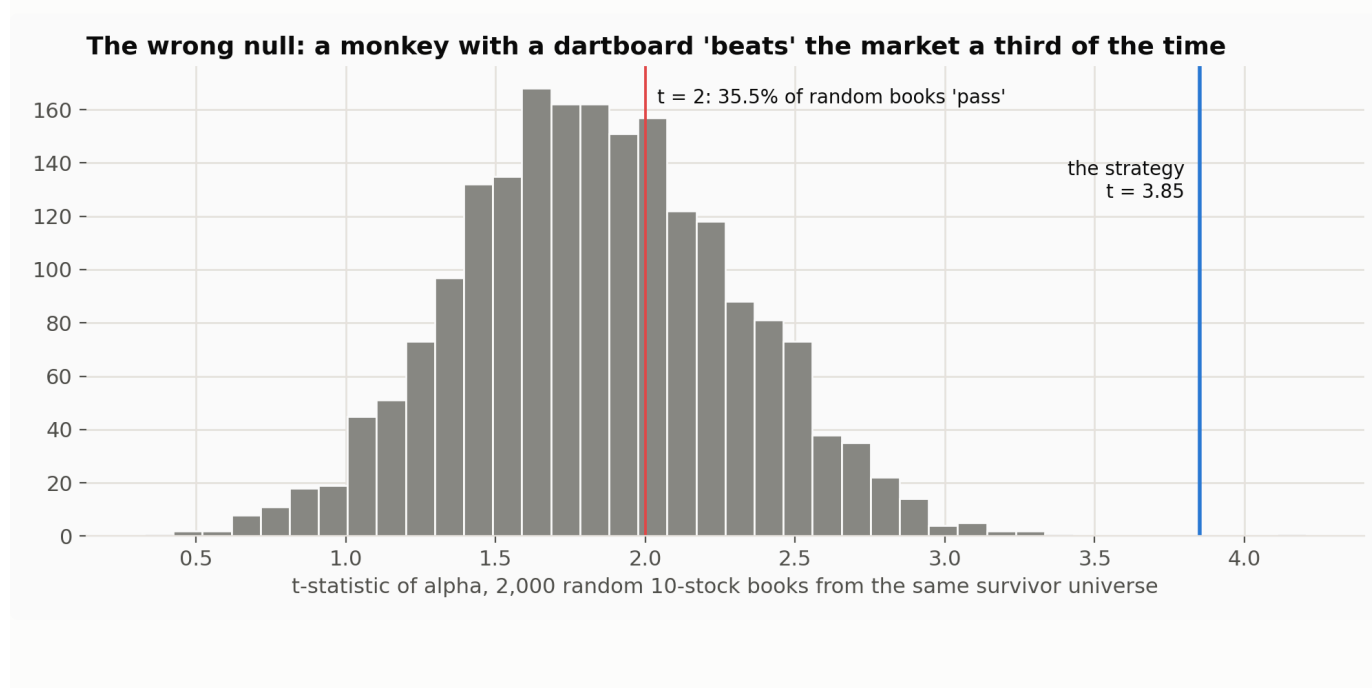
| Universe           | Sharpe on today's members | Sharpe on members at the time | Overstatement      | Method                                             |
|--------------------|---------------------------|-------------------------------|--------------------|----------------------------------------------------|
| US (S&P 500)       | 1.267                     | 0.794                         | +60% (haircut 37%) | today's members vs point-in-time members, Sharadar |
| Europe (STOXX 600) | 0.758                     | 0.529                         | +43% (haircut 30%) | same panel, point-in-time mask off vs on           |
| UK                 | 0.865                     | 0.694                         | +25% (haircut 20%) | current members vs point-in-time                   |
| Denmark            | 0.973                     | 0.683                         | +43% (haircut 30%) | current members vs point-in-time; small, noisy     |



The haircut ranges from 20% to 37% of the Sharpe ratio. The effect is even larger on a placebo test, which replays the strategy's exact holding schedule with randomly chosen names. On today's index members the strategy looked extraordinary ( $z = 4.92$ ). On members at the time it sits at the 91th percentile ( $z = 1.34$ ): about one random portfolio in eleven did better. Survivorship and momentum interact. Among survivors, past winners are disproportionately the names that kept winning, so the same test on the convenient universe inflates the answer about 3.7 times.

## 4. The wrong null: a dartboard beats the market

A common way to judge a stock-picking strategy is to ask whether its alpha is significantly above zero. But zero is the wrong benchmark when the universe itself is biased. I drew 2,000 random ten-stock books from the same survivor universe and measured their alpha the same way.



The random books averaged a  $t$  of 1.84 and 35.5% of them cleared  $t > 2$ . The strategy's own  $t$  of 3.85 beat almost all of them (0.05% did better). That test was run on the frozen, bugged book, and it shows how much apparent skill the universe hands out for free. **Lesson: test against random portfolios drawn from the same universe, not against zero.**

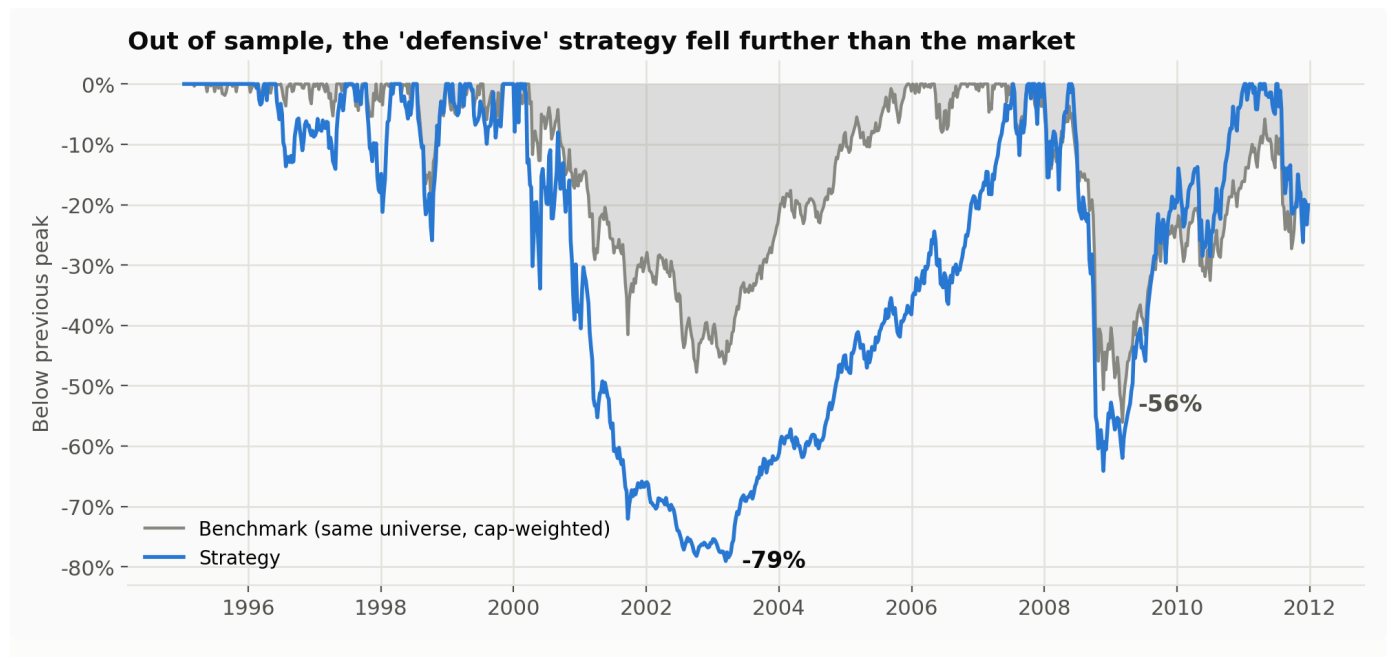
## 5. Out of sample: sixteen years the strategy had never seen

Before looking, I registered one test. The same rules would run on a point-in-time database of European and American large caps, 1995–2011, a period the strategy had never touched. The pass rule was  $t > 2$ .

**Table 3. The out-of-sample test**

|                                         | Strategy                                | Benchmark |
|-----------------------------------------|-----------------------------------------|-----------|
| Period, weeks                           | 1995-01-13 to 2011-12-23, 885           | same      |
| CAGR                                    | +9.6%                                   | +8.6%     |
| Sharpe                                  | 0.475                                   | 0.540     |
| Maximum drawdown                        | <b>-79.1%</b>                           | -56.1%    |
| Beta to benchmark                       | 1.18                                    | 1.00      |
| Alpha / yr, 95% CI (record)             | <b>+1.15%</b> [-8.1%, +10.6%], $t$ 0.24 |           |
| Alpha / yr, plain OLS (re-derived here) | +1.15%, $t$ 0.29                        |           |
| Europe leg alone                        | +0.31%, $t$ 0.05                        |           |
| America leg alone                       | +2.30%, $t$ 0.34                        |           |

*Benchmark: cap-weighted return of the same point-in-time universe (top 600 Europe, top 500 America), equal-weighted across the two regions, as pre-registered. The strategy ends with more wealth than the benchmark only because it carried more market risk (beta 1.18); per unit of risk it lost.*



It did not replicate. The in-sample drawdown advantage also disappeared. The 2012–2026 window contains no full bear market, and 1995–2011 contains two. A strategy that looked defensive in-sample lost 79% peak to trough, far more than the market.

## 6. Counting the trials

Every choice you make after seeing the data (a feature, a lookback, a region, a parameter) is another draw from the lottery. This project counted them: 33 pre-registered trials by the time of the last correction.

**Table 4. What  $t$  is needed when you have tried  $N$  things**

| Number of strategies tried | $t$ needed (Bonferroni, 5% two-sided) | Does $t = 2.90$ clear it? |
|----------------------------|---------------------------------------|---------------------------|
| 1                          | 1.96                                  | yes                       |
| 5                          | 2.58                                  | yes                       |
| 13                         | 2.89                                  | yes                       |
| 24                         | 3.08                                  | no                        |
| 33                         | 3.17                                  | no                        |
| 100                        | 3.48                                  | no                        |
| 235                        | 3.70                                  | no                        |

The first  $t$  of 2.90 would have survived only up to 13 trials. The Harvey, Liu & Zhu (2016) Bonferroni haircut at 33 trials takes the  $t$  to 1.54, a 47% haircut to the Sharpe ratio, before any of the corrections above. The deflated Sharpe ratio of the best of the first 24 variants (Bailey & López de Prado, 2014) was

0.9359, below the 0.95 threshold. **Lesson: write the count down before you start, and raise the bar with it.**

## 7. How typical is this? Relevance statistics

**Table 5. This strategy's haircut against the literature**

| Study                          | Sample                                        | Finding                                                            | Haircut    |
|--------------------------------|-----------------------------------------------|--------------------------------------------------------------------|------------|
| McLean & Pontiff (2016)        | 97 published anomalies                        | Returns lower out of sample (before publication)                   | 26%        |
| McLean & Pontiff (2016)        | 97 published anomalies                        | Returns lower after publication                                    | 58%        |
| Suhonen, Lennkh & Perez (2017) | 215 alternative-beta strategies from 15 banks | Median Sharpe decline from backtest to live                        | 73%        |
| Hou, Xue & Zhang (2020)        | 452 anomalies                                 | Share failing $ t  \geq 1.96$ once microcaps are controlled        | 65%        |
| Hou, Xue & Zhang (2020)        | 452 anomalies                                 | Share failing a multiple-testing hurdle of $t \geq 2.78$           | 82%        |
| Wiecki et al. (2016)           | 888 algorithms on Quantopian                  | Backtest Sharpe explains less than 2.5% of live Sharpe (R-squared) |            |
| <b>This strategy</b>           | one momentum book, 2012–2026                  | alpha, first result → number of record                             | <b>79%</b> |
| <b>This strategy</b>           | same                                          | alpha, first result → out of sample                                | <b>77%</b> |
| <b>This strategy</b>           | same                                          | Sharpe, first result → number of record                            | <b>30%</b> |

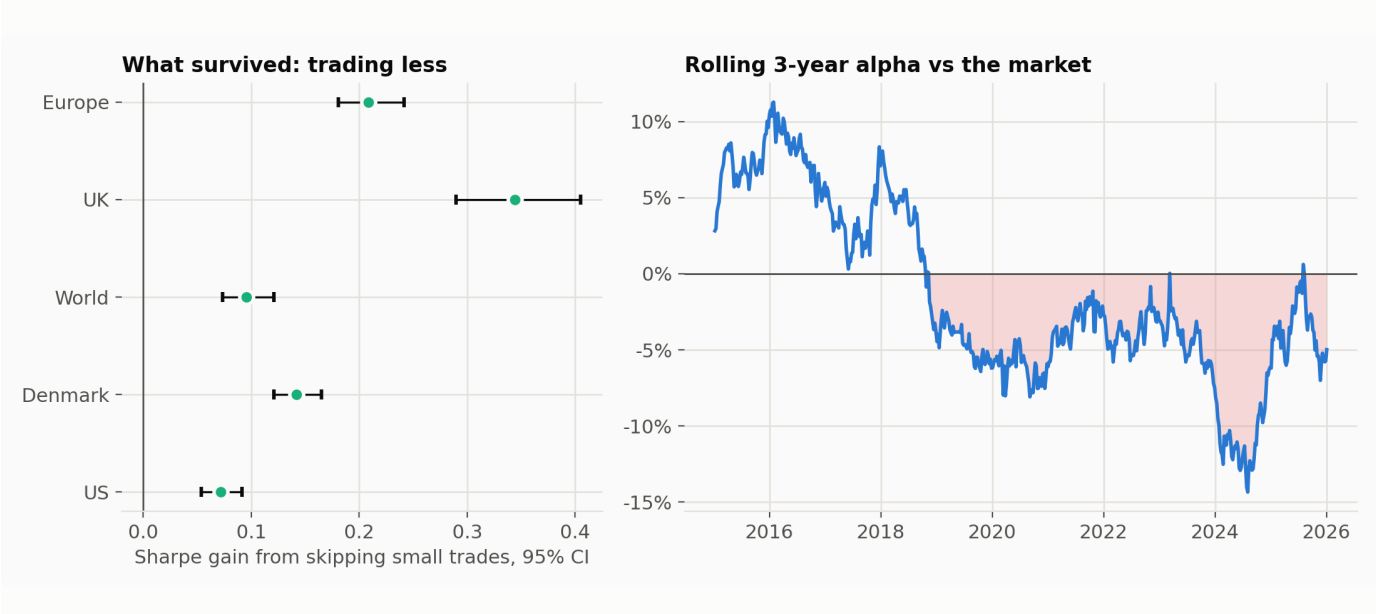
A 79% alpha haircut sounds dramatic. It is ordinary. McLean & Pontiff find published anomalies lose 58% of their return after publication. Suhonen, Lennkh & Perez find bank-marketed alternative-beta strategies lose a median 73% of their backtested Sharpe once live. Hou, Xue & Zhang fail most of 452 anomalies on a clean replication. Wiecki et al. find that a backtest's Sharpe barely predicts its live Sharpe. On that evidence, **the right prior for any backtest is that most of it will go**. This one went slightly faster than average, because each correction here was cheap to make.

## 8. What survived: trading less

One result cleared every test I put to it. The strategy skips trades below a fixed size (a no-trade band), and that measurably improves its Sharpe ratio in every region.

**Table 6. The no-trade band**

| Region                        | Sharpe gain from the band | 95% CI (paired block bootstrap) |
|-------------------------------|---------------------------|---------------------------------|
| Europe                        | +0.209                    | [+0.181, +0.242]                |
| UK                            | +0.344                    | [+0.290, +0.405]                |
| World                         | +0.096                    | [+0.073, +0.121]                |
| Denmark                       | +0.142                    | [+0.121, +0.165]                |
| US                            | +0.071                    | [+0.054, +0.091]                |
| <b>Book (50/50 US/Europe)</b> | <b>+0.160</b>             | <b>[+0.137, +0.186]</b>         |



It is not selection skill. It is cost control: the same strategy, run with less churn. Compared with an index fund, the strategy still has nothing to show. Its rolling three-year alpha has been negative in 99% of weeks since 2019 and stood at -5.0% at the last reading (2026-01-02).

## 9. A checklist before you believe a backtest

1. **Same return basis** for every leg and the benchmark (total return, same currency).
2. **Point-in-time universe:** the index as it was, including the names that later died.
3. **Clean data:** one bad split (a 1:1000 reverse split read as +107,385% in a week) was held by this momentum strategy for 63 weeks.
4. **The right null:** random portfolios from the same universe, not zero.
5. **Count every trial** and raise the t-bar with it; report the deflated Sharpe.
6. **Test on untouched data** with a rule written down before looking.
7. **Report the drawdown path**, not only the Sharpe, and a sample with a bear market in it.

## 10. Caveats

- This is one strategy. The corrections are general, but their sizes are not.
- The point-in-time European universe is itself incomplete (26% of historical names lack prices), so its survivorship estimate is a lower bound.
- The out-of-sample period (1995–2011) comes from a licensed database. Only summary statistics and charts are published, not the series.
- Currency restatement was measured on the pre-fix book (row 3 of Table 1).

### Want this run on your market, idea or strategy?

I run bespoke versions of these studies: any strategy here on the market or universe you choose, a signal or idea you propose, tested under the same rules (hypothesis and pass mark fixed before the data are seen), or a review of an existing strategy's returns, costs, factor exposures and how its parts interact.

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## Reproduce

```
python code/analyse.py          # re-derives the numbers of record, writes results/summary.json
python code/make_figures.py     # figures/
python code/build_article.py    # this article and the LinkedIn teaser
python code/make_pdfs.py       # PDFs
```

The weekly return series live in `data_private/` and are not published: they derive from licensed data. Everything else (summary files, sources for each number, code) is in the repository.

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*Not investment advice.*